Kannan Sekar Annu Radha

London, UK | kannansekara@gmail.com | linkedin.com/in/kannan-sekar | github.com/kannansa | sakannan.com

Summary

Software Engineer and Systems/ML Researcher (Physics & CS) targeting **SWE and SRE** roles. Experienced in scalable backend services, distributed **Linux/HPC** compute pipelines, and low-latency **CUDA/ML** systems, with end-to-end production ownership across containerisation (**Docker, Kubernetes, GCP**), automated **CI/CD**, observability, and latency/throughput optimisation.

Technical Skills

Languages: Python, C, C++, Go, Java, Swift, Objective-C, Bash/Shell, Haskell
Systems, Cloud & SRE: Linux/Unix Internals, Docker, Kubernetes, GCP, CI/CD Automation, HPC/Slurm, CUDA/GPU Optimisation, Git
Machine Learning & Data: PyTorch, TensorFlow, CoreML, Scikit-learn, NumPy, Pandas, LAMMPS, OVITO, NetworkX
Architecture & Delivery: Distributed Systems, REST/GraphQL APIs, Microservices, Observability/Telemetry, Latency & Memory Profiling

Experience

Galleries Medical Practice — London, UK
Software Engineer (Backend, Cloud & Client Systems) — *Nov 2025 – Present*

- Architected and deployed production web platforms, REST backend services, and chatbot workflows (globaltravelvaccinations.com) on **GCP and Linux**, maintaining high availability across clinical user journeys.
- Engineered end-to-end features across client, API, and cloud layers; improved operational readiness and incident triage via structured telemetry, logging hooks, and automated health checks.
- Partnered with stakeholders to translate ambiguous requirements into fault-tolerant architectures, automated regression suites, and zero-downtime releases.

King's College London (Dept. of Physics) — London, UK
Research Fellow / KURF Fellow (Distributed HPC & Applied ML) — *May 2025 – Nov 2025*

- Architected physics-informed ML models in **Python and C++** for carbon nanocluster simulations, improving predictive accuracy by **80%** and cutting simulation runtime by **60%**.
- Engineered high-throughput GPU data ingestion and training pipelines (**TensorFlow, CUDA**), reducing I/O and preprocessing latency by **3x** across large-scale structural datasets.
- Orchestrated distributed **LAMMPS** workloads on **Linux HPC clusters**, automating batch scheduling, failure recovery, and analysis across **5,000+ atomic trajectories** (**OVITO**).
- Applied unsupervised representation learning (**PCA, t-SNE**) to automate outlier detection and validate predicted vs. simulated energy distributions at scale.

Kannan Industrials — London, UK
Founder & Lead Software Engineer — *Feb 2021 – Present*

- Designed, built, and operated 4 production apps (*1minuteDOEShelp, iPong, DabCounter, OK Gym*) and cloud web services (sakannan.com), owning end-to-end architecture, deployment, and reliability.
- Optimised on-device **CoreML** and signal-processing pipelines via memory and concurrency profiling, boosting real-time inference throughput by **70%** under strict hardware constraints.
- Engineered automated build, test, and release **CI/CD pipelines** alongside modular service architectures, cutting deployment lead time by **50%** and eliminating manual release toil.

NHS Digital — Leeds, UK
Data Science Intern — *Nov 2019 – Dec 2019*

- Built and tuned a clinical NLP classification pipeline (**Python, Word2Vec, LSTM**) to predict ICD-9 codes from unstructured text, increasing validation accuracy from **42%** $\rightarrow$ **71%**.
- Developed **t-SNE** anomaly-detection workflows for rare diagnoses and packaged reproducible training pipelines for handover.

Kennedy Institute of Rheumatology (University of Oxford) — Oxford, UK
Machine Learning Research Intern — *Nov 2019*

- Engineered deep CNN architectures for histopathology and biomedical imaging benchmarks (MNIST, CIFAR), achieving a **40% accuracy improvement** over baseline models.
- Implemented an LSTM reinforcement learning trading system, boosting predictive accuracy by **70%** and cutting GPU training time by **30%** via mixed-precision (**FP16/CUDA**) optimisation.

Snap Inc. — Gateshead, UK
AR Lens Developer — *Sep 2018 – Aug 2019*

- Engineered real-time computer vision and AR experiences reaching **2.5M+ views** and **150K+ downloads**, profiling shader execution, frame-time latency, and memory footprints on mobile hardware.

Selected Systems Projects

HPC Nanocluster Compute Pipeline | *C++, CUDA, Python, Linux, LAMMPS*

- Built parallel batch-processing and GPU ML analysis pipelines on Linux, automating dataset ingestion, fault-tolerant job execution, and high-dimensional visualisation.

Real-Time Edge RL & Motion Engine (iPong / DabCounter) | *Swift, C++, CoreML, Concurrency*

- Engineered a latency-aware on-device RL agent and high-frequency sensor classification pipeline achieving $<$**50ms** end-to-end gesture recognition latency.

Education

King's College London, UK – BSc Physics (**Alessandro de Vita Computational Physics Prize Winner**) — *2024 – 2028*
University College London (UCL), UK – MEng Computer Science — *2021 – 2023*
Royal Grammar School, Newcastle, UK – A-Levels: **A*A*A*** (Mathematics, Further Mathematics, Physics) — *2017 – 2020*